



IRIDIUM NEXT

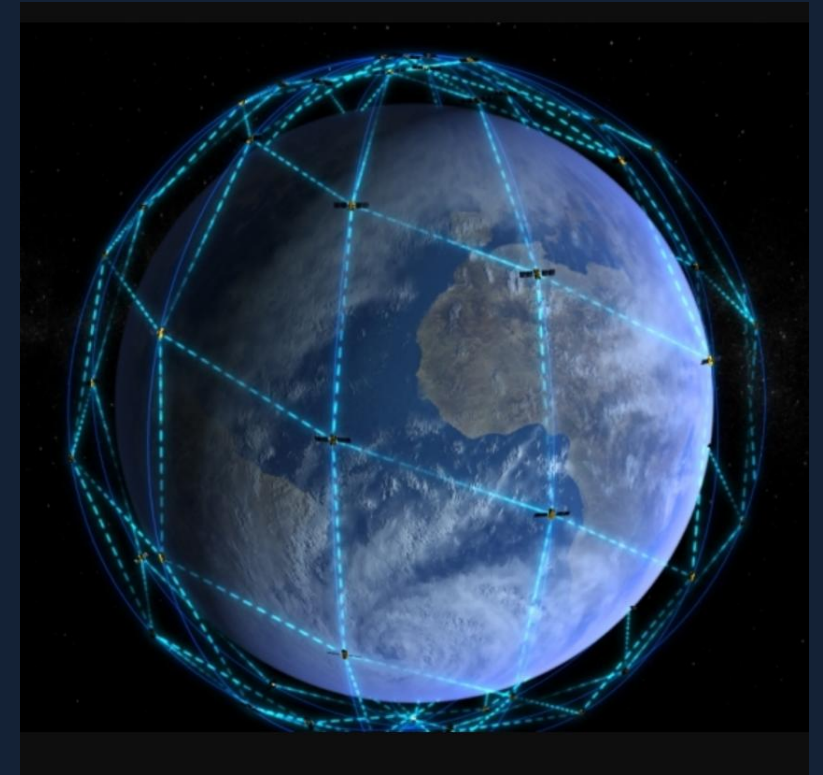
BY IRIDIUM COMMUNICATIONS INC.

William Walsh, Chinmaya Salinamakki, Jenna Wilsher

ECE 5610 Satellite Communication Systems 5/6/26

BACKGROUND

- Iridium NEXT is a LEO constellation of 66 satellites launched with the intent to solve the problem of complete global communication for voice, data, and IoT
- Iridium NEXT is a successor to the Iridium system, switched over 2017-2019 involving 8 individual launches to replace all 66
- Transition is said to have cost \$3 billion dollars
- Network access is available through devices with a compatible satellite terminal that operates on the L-band frequency and airtime plan
- Industries who use Iridium NEXT:
 - o Maritime/Aviation
 - o Emergency Response
 - o Government/Military



Artist depiction of Iridium NEXT constellation



ORBITAL CONFIGURATION

- Number of Satellites: 66 + 9 redundant satellites
- Number of Orbital Planes: 6
- Separation of Orbital Planes: 30°
- Inclination of Orbital Planes: 86.4°
- Orbit Height: 780km (LEO Orbit)
- Orbital Velocity: 27000km/hr
- Orbit Period: 100 minutes
- Satellites Launched by Space-X Falcon 9 Rockets
- Chemical Propulsion (Hydrazine Tank) used to bring orbit up from 625km to 780km and make orbital corrections



SUBSYSTEMS

1. AOCS – Attitude and Orbit Control System
 - Attitude sensing: star trackers, magnetometer
2. Propulsion System – Catalyst Thrusters
3. Communications System
4. TT&C – Telemetry, tracking, and control
 - Omnidirectional antennas
 - Data is sent to Space Network Operations Center in Leesburg, Va
5. Power System – 2 kW solar power
6. Hosted Payloads



Space Network Operations Center,
Leesburg, Va

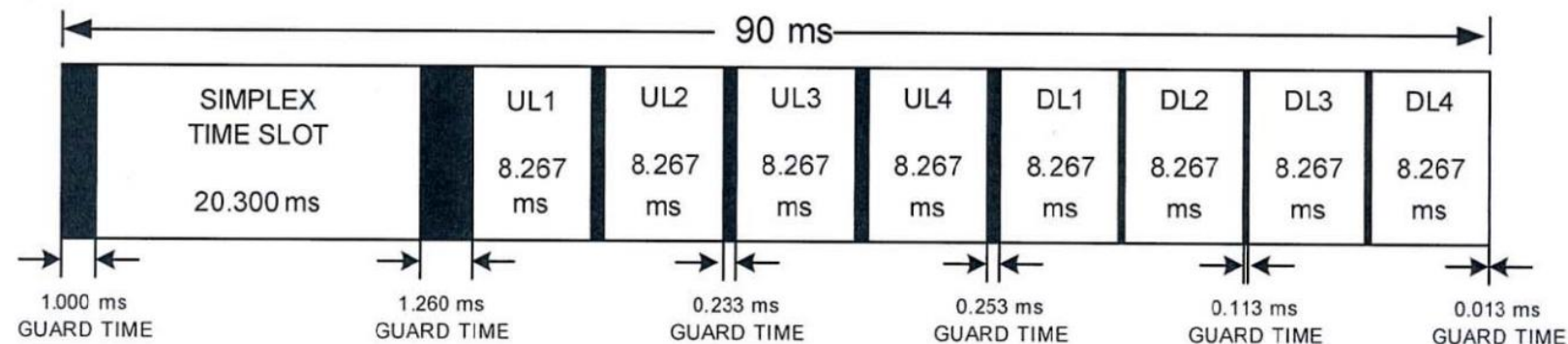
Use	Fuel Mass
Insertion	38.0 kg
Drift	71.6 kg
Station keeping	5.6 kg
Deorbit	39.6 kg
Residuals	9.3 kg
Total	164.0 kg



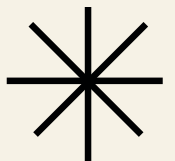
Leonardo Star
Tracker

COMMUNICATIONS OVERVIEW

Link Name	Purpose	Frequencies	Antenna	Rationale
Service Link	- Interface b/w satellite and user devices - Responsible for most of satellite profit	L-Band, 1616.0-1626.5 MHz, Time Division Duplex (TDD)	48-beam Tx/Rx Phased Array (sat), small user antennas	Low attenuation -> High reliability for mobile users
Feeder Link	- Connection between satellite and ground gateway	Ka-Band, Up: 29.1-29.3 GHz Dn: 19.4-19.6 GHz	High-gain dish antenna	High Bandwidth
Intersatellite Link	- Routes information b/w satellite and four adjacent satellites in the forward, aft, and neighboring planes	K-Band, 22.18-22.38 GHz	Highly directional	Allows for routing in space -> No ground dependence



TDMA Frame



LINK BUDGET ANALYSIS OF FEEDER LINK

- Operates at 29.1-29.3GHz on Uplink
- Operates at 19.4-19.6GHz on Downlink
- 13 Transponders/Channels with a Bandwidth of 14MHz
- Link Budget Formula: $Pr = Pt + Gt + Gr - Lp - Ltot$

TRANSMIT AND NOISE POWER AND ANTENNA GAIN

Uplink

- Ground Station EIRP $EIRP = P_t + G_t = 67dBW$
- Satellite Receive Gain $G_r = 30.1dBi$
- Satellite Receiver Noise Temperature $T = 1295K$
- Noise/Channel Bandwidth $B_n = 14MHz$
- Satellite Receiver Noise Power $N = kTB_n = -126dBW$

Downlink

- Assume $P_t = 10dBW$
- $D = 0.1 - 0.3m, \eta = 0.6$ for satellite antenna
- $\lambda = \frac{c}{f} = \frac{3(10)^8}{19.5GHz} = 0.015385m$
- Satellite Transmit Gain $G_t = \eta \frac{(\pi D)^2}{(\lambda^2)} = 26.5dBi$
- $D = 3 - 4m, \eta = 0.6$ for ground antenna
- Ground Station Receive Gain $G_r = \eta \frac{(\pi D)^2}{(\lambda^2)} = 53.2dBi$
- Ground Station Noise Temperature $T = 731K$
- Ground Station Noise Power $N = kTB_n = -128.5dBW$

PATH LOSS AND TOTAL LOSS

Uplink

- When Satellite is 8.2 degrees above horizon to ground station is the maximum distance/path loss
- Orbit Height $H = 780km$
- $d = R_E \left[\sqrt{\frac{(H+R_E)}{R_E} - \cos(\varphi)^2} - \sin(\varphi) \right] = 2463km$
- $L_p = \left(\frac{4\pi df}{c} \right)^2 = \left(\frac{4\pi(2463km)(29.2GHz)}{3(10)^8} \right)^2 = 189.6dB$
- Main Source of Additional Loss is Rain Attenuation for the Ka Band. Consider for Ground Stations in Dry Climates
- $L_{tot} \approx 11dB$

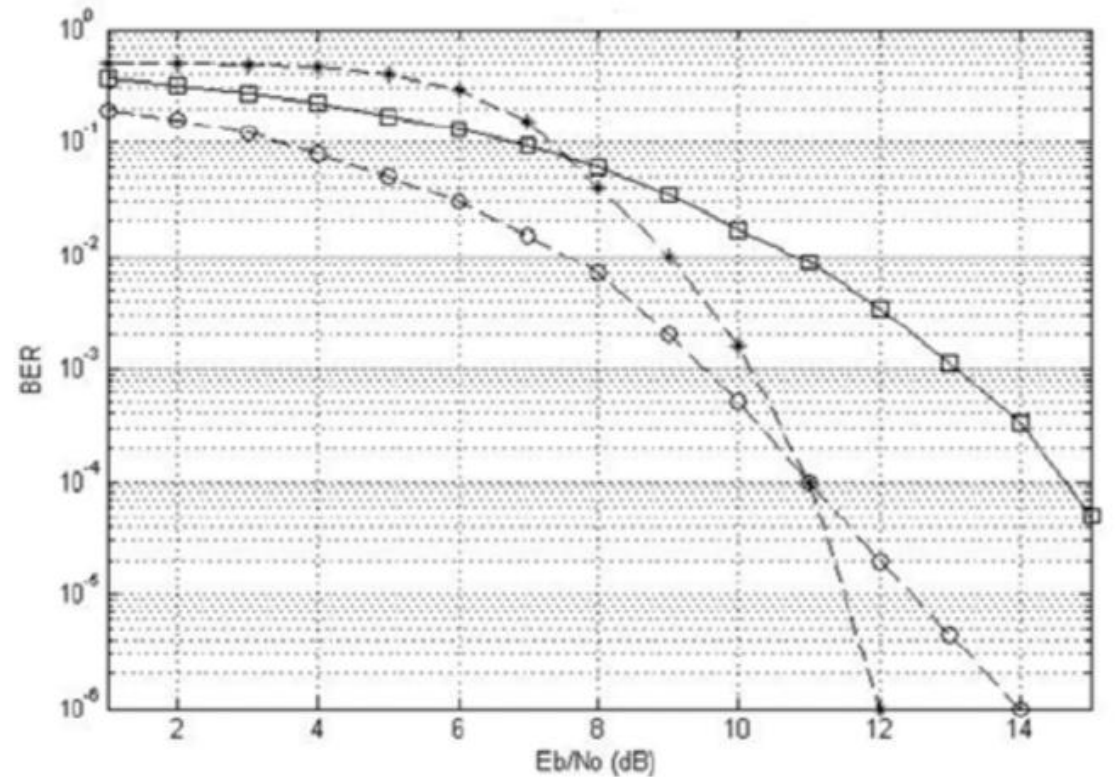
Downlink

- Same maximum distance as uplink, different frequency
- $L_p = \left(\frac{4\pi df}{c} \right)^2 = \left(\frac{4\pi(2463km)(19.5GHz)}{3(10)^8} \right)^2 = 186dB$
- $L_{tot} \approx 11dB$

REQUIRED C/N

Uplink/Downlink

- Assume bit error rate of 10^{-6} .
- QPSK with Convolutional Coding
- $\frac{Eb}{No} = 12dB$
- $B = 14MHz$
- $Rb = \eta B \approx 25Mbps$
- $\frac{C}{N} = \frac{Eb}{No} + 10\log_{10}\left(\frac{Rb}{B}\right) = 14.5dB$



Effect of coding: solid line (squares) = QPSK without coding; dashed line (circles) = QPSK with Hamming (7, 4), $t = 1$ coding; dashed line (asterisks) = QPSK with Convolutional coding (rate $3/4$)

RESULTS

Uplink

Parameter	Value
Effective Isotropic Radiated Power	67dBW
Satellite Antenna Receive Gain	30dBi
Path Loss	189.6dB
Other Loss Sources	11dB
Power Received	106.5dBW
Receiver Bandwidth	14MHz
Noise Power	-126dB
C/N	19.5dB
C/N Required	14.5dB
Link Margin	4dB

Downlink

Parameter	Value
Satellite Transmit Power	10dBW
Satellite Antenna Transmit Gain	26.5dBi
Ground Station Antenna Receive Gain	53.2dBi
Path Loss	186dB
Other Loss Sources	11dB
Power Received	107.3dBW
Receiver Bandwidth	14MHz
Noise Power	-128.5dB
C/N	21.2dB
C/N Required	14.5dB
Link Margin	6.7dB

HOSTED PAYLOADS

Payload Stats:

- 50kg of additional sensor weight

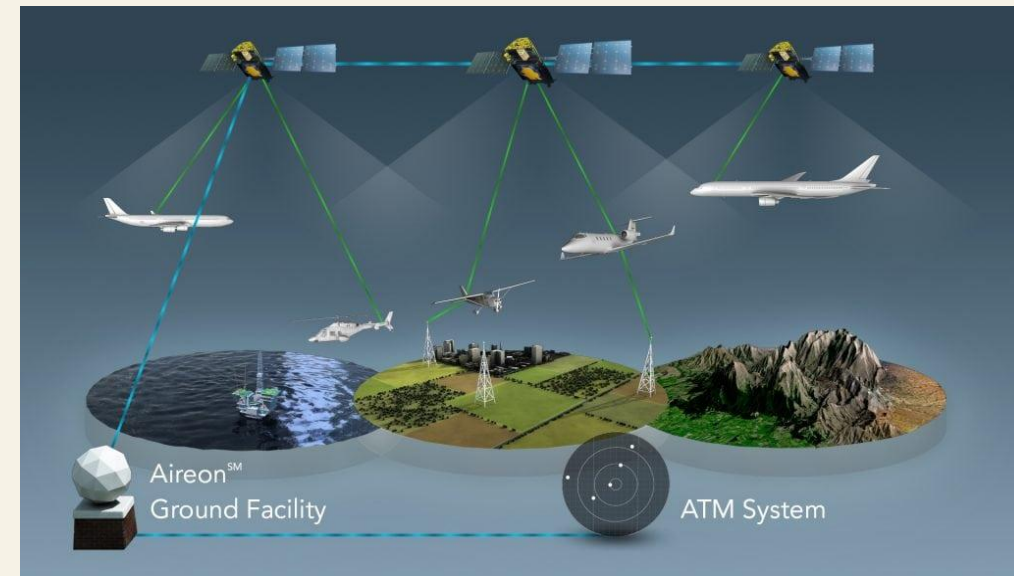
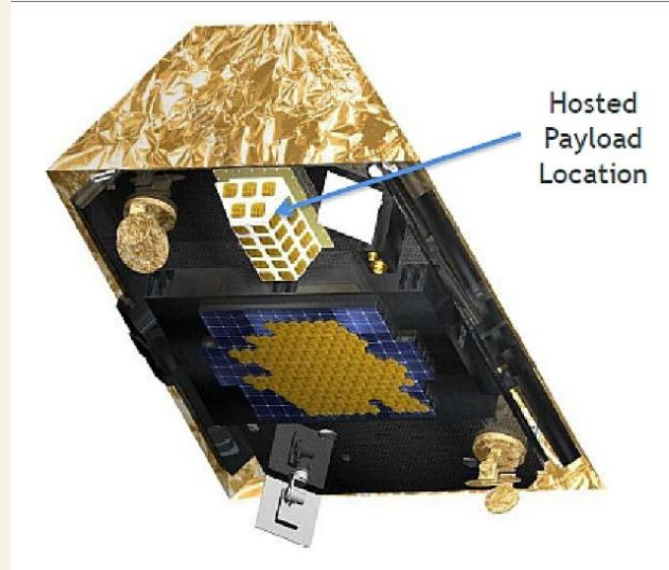
Note: Each Iridium NEXT satellite weights about 860 kg in total.

- 50 W average power
- 40 cm X 70 cm X 30 cm

Primary Contributors:

1. Aireon LLC: Automatic Dependent Surveillance-Broadcast (ADS-B)
 - Aids aircraft in broadcasting their position, flight data, Aircraft ID, etc for traffic control/collision avoidance
2. Harris Corporation: Maritime Automatic Identification System (AIS)
 - Relies on maritime vehicles broadcasting their GPS derived position, velocity, ID, etc for traffic control

Hosted Payload
location on
Iridium NEXT
body.



Aireon Detection Strategy

Sensor	Latency	# Satellites	Freq.
ADS-B	1.5 seconds	66/66	112-bit, 1090 MHz
AIS	< 1 minute	58/66	VHF Maritime Band, 162 MHz

FUTURE OF IRIDIUM

Q: What makes Iridium-NEXT relevant?

A: High reliability, global communications

Q: Can it compete with newer systems?

A: Yes, but not in raw data rate

Q: How long will Iridium-NEXT be useful?

A: Projected lifespan through mid-2030s

Q: Will Iridium-NEXT be replaced?

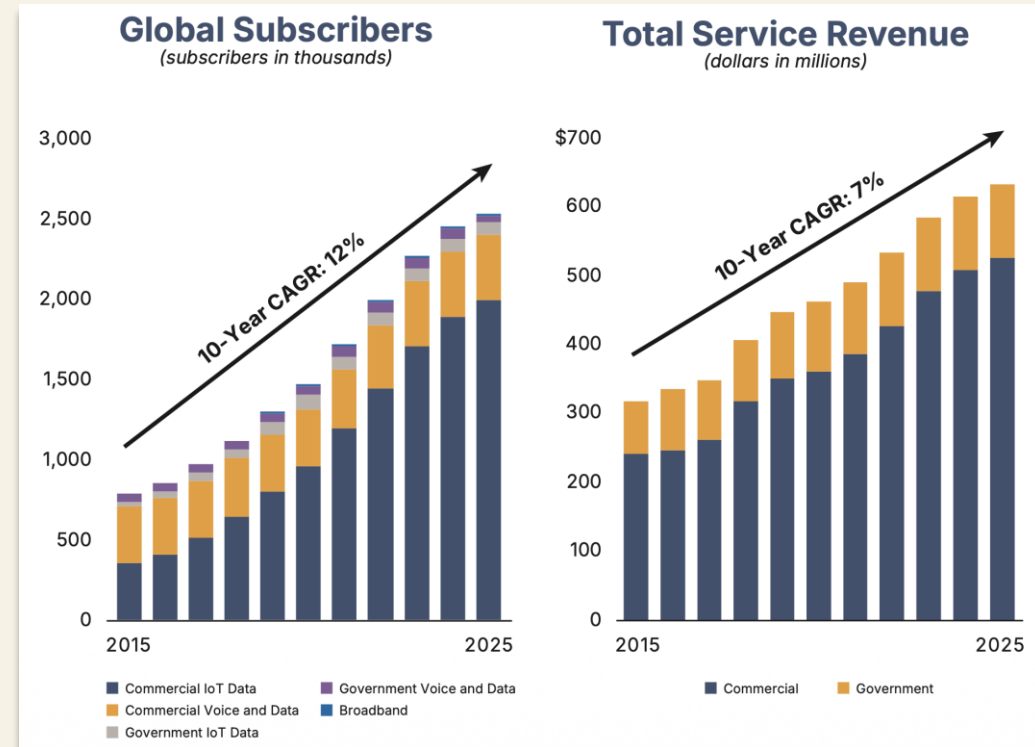
A: No public plans for replacement



Iridium Certus 9704



RockBLOCK APNT



THANKS FOR LISTENING!

Image References:

[1] *Iridium® NEXT constellation, built by Thales Alenia Space, now completely deployed in orbit.*

Thalesgroup.com. (n.d.). <https://www.thalesgroup.com/en/news-centre/press-releases/iridiumr-next-constellation-built-thales-alenia-space-now-completely>

[2] Writer, S. (2017, June 26). *AIREON launches, deploys 10 more Iridium NEXT satellites with ADS-B payloads.* Aviation Tech Today. <https://www.aviationtoday.com/2017/06/26/aireon-launches-deploys-10-iridium-next-satellites-ads-b-payloads/>

[3] *Iridium-next.* Gunter's Space Page. (n.d.-b). https://space.skyrocket.de/doc_sdat/iridium-next.htm

[4] SatCatalog. (n.d.). *AA-STR.* Retrieved May 6, 2026, from <https://www.satcatalog.com/component/aa-str/>

[5] Iridium Communications Inc. (2026). *2025 annual report.* https://filecache.investorroom.com/mr5ir_iridium/903/Iridium_Communications_Inc_2025_Annual_Report.pdf

[6] Ground Control. (n.d.). *RockBLOCK APNT evaluation kit.* Retrieved May 6, 2026, from <https://www.groundcontrol.com/product/rockblock-apnt/>

[7] Iridium Communications Inc. (n.d.). *Iridium Certus™ 9704 satellite IoT developer kit.* Retrieved May 6, 2026, from <https://www.iridium.com/products/iridium-certus-9704-satellite-iot-developer-kit>

